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Association Between Collision Sport Career Duration and Gait Performance in Male Collegiate Student-Athletes

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Background: Investigations of estimated age of first exposure to repetitive head impacts from collision and contact sports have shown no associations with neurocognitive or neurobehavioral function at the collegiate level, but the effect of career duration may be a more comprehensive factor. Understanding whether longer career duration influences gait performance would provide insights into potential neurological impairment.

Purpose: To examine the relationship between career duration of collision sports and single/dual-task gait performance in collegiate student-athletes.

Study Design: Cross-sectional study; Level of evidence, 3.

Methods: We recruited 168 male student-athletes from collision sports: football, lacrosse, ice hockey, and wrestling (mean ± SD age, 19.2 ± 1.3 years; height, 184.5 ± 7.2 cm; mass, 94.3 ± 15.9 kg; estimated age of first exposure, 8.6 ± 3.1 years; career duration, 10.6 ± 3.0 years). All participants completed a baseline single- and dual-task gait assessment before the start of their athletic season. Inertial measurement units were used to measure gait speed and stride length. During the dual task, participants were asked to perform working memory cognitive tasks while walking. The dependent variables were single/dual-task gait speed and stride length, cognitive accuracy, and dual-task cost. The relationship between career duration, analyzed as a continuous variable, and the dependent variables was analyzed using a linear regression.

Results: There were no significant associations between career duration and single-task gait speed (1.16 ± 0.16 m/s; $\beta = -0.004$; $P = .35$; 95% CI = −0.012 to 0.004; $\eta^2 = 0.005$) or dual-task gait speed (1.02 ± 0.17 m/s; $\beta = -0.003$; $P = .57$; 95% CI = −0.011 to 0.006; $\eta^2 = 0.002$). There were also no significant associations between career duration and single/dual-task stride length, cognitive accuracy, or dual-task cost.

Conclusion: Career duration among collegiate collision sport athletes was not associated with single- or dual-task gait performance, suggesting that a greater exposure to repetitive head impacts is not detrimental to dynamic postural control at the college level. However, the effects of diminished gait performance over the lifetime remain to be elucidated.

Keywords: head injuries/concussion; gait analysis; repetitive head impacts; dual task

More than 280,000 male student-athletes compete in National Collegiate Athletic Association (NCAA) sports, and approximately 35% of those individuals participate in collision sports (eg, football, men's ice hockey).[24] When compared with other sport athletes, collision sport athletes are likely to experience more repetitive head impacts (RHIs) because of the nature of collision sport rules, and there is growing concern surrounding these RHIs and resulting effects on long-term brain health.[26,34] While the acute consequences of concussion are better understood,[6,33] the RHIs that do not manifest in the signs and symptoms for concussion diagnosis may also have short- and long-term repercussions.[3,4]

RHIs occur frequently in collision sports, and by the end of a college career, a football lineman could sustain thousands of head impacts.[39] Additionally, earlier exposure via tackle football may be associated with later life neurocognitive and neurobehavioral impairments, as well as macrostructural/microstructural alterations, but not chronic traumatic encephalopathy pathological severity.[1,44,47,48] At the collegiate level, recent investigations of healthy NCAA student-athletes found no evidence of an association between estimated age of first exposure (eAFE) and worse

The American Journal of Sports Medicine
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DOI: 10.1177/03635465221104685
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outcomes on clinical concussion measures, including neurocognitive function, psychological distress, experienced symptoms, or symptom recovery.[11-14,25] The relationship between eAFE and functional outcomes has been robustly examined in recent years; however, overall collision sport career duration, as opposed to eAFE, has been recently proposed as the primary diagnostic exposure variable for traumatic encephalopathy syndrome.[26] Overall career duration of collision sports participation may provide a more direct measure of the role of RHIs on functional outcomes versus eAFE, which does not account for number of years of participation. Furthermore, career duration is a more robust predictor of long-term health.[26] There is minimal but mixed literature investigating the role of collision sport career duration on long-term outcomes, and the existing studies have largely been conducted on retired or deceased former athletes.[15,36,37,45]

No association between earlier eAFE and postural stability has been observed in collegiate athletes[11,13]; however, the existing investigations used the Balance Error Scoring System, which, despite being commonly used by NCAA athletic trainers,[27] is limited by poor psychometrics and measures postural stability in a quiet stance domain.[7] Conversely, gait is a dynamic postural control task that serves as an element of neurological health.[17,43] Gait speed is considered to be a psychometrically sound indicator of physiological changes and future health status, and slower gait speeds reflect a conservative gait strategy in postconcussion athletic populations.[8-10,32,40] Gait speed also predicts functional decline in older adults, and midlife gait speed may be associated with accelerated aging, compromised brain health, and neurocognitive function.[43] Furthermore, dual-task gait represents an important component of neurological health, as it assesses the motor and cognitive domains simultaneously.[17,42,49] Dual-task gait alterations have been observed in retired professional football players, although this finding was independent of career duration[31] and the translation to a collegiate population is unknown.

Investigations of eAFE to RHI from collision and contact sports have not yielded significant associations with neurocognitive or neurobehavioral function at the collegiate level. However, it is unknown whether longer career duration influences gait performance in a collegiate athlete population, which, if found, could suggest that RHIs play a significant role in gait-based central nervous system health. Therefore, the purpose of this investigation was to examine the relationship between estimated career duration of collision sport participation and single/dual-task gait performance in collegiate male student-athletes. We hypothesized that a longer career duration would not impair gait performance in a collegiate athlete population.

METHODS

Participants

Participants were recruited from 2 NCAA Division I athletic institutions between 2015 and 2019. The inclusion criteria were male student-athletes participating in an intercollegiate collision sport (football, men's lacrosse, men's ice hockey, or wrestling) consistent with previous literature whereby collision sports were defined as those involving purposeful, legal body-to-body collisions.[35] In accordance with the traumatic encephalopathy syndrome diagnostic criteria, our primary exclusion criteria included a career duration <5 years.[26] This exclusion was applied

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Opinions, interpretations, conclusions and recommendations are those of the authors and are not necessarily endorsed by the Department of Defense.

Submitted September 13, 2021; accepted May 5, 2022.

One or more of the authors has declared the following potential conflict of interest or source of funding: This publication was made possible with support from the Grand Alliance CARE Consortium (Concussion Assessment, Research, and Education), funded in part by the National Collegiate Athletic Association and the Department of Defense. The US Army Medical Research Acquisition Activity is the awarding and administering acquisition office. This work was supported by the Office of the Assistant Secretary of Defense for Health Affairs through the Combat Casualty Care Research Program and endorsed by the Department of Defense through the Joint Program Committee 6/Combat Casualty Care Research Program–Psychological Health and Traumatic Brain Injury Program (W81XWH1420151). Research reported in this publication was also supported by the National Institute of Neurological Disorders and Stroke of the National Institutes of Health (NIH) (R03NS104371 and R01NS100952). D.R.H. has received research support from the Eunice Kennedy Shriver National Institute of Child Health and Human Development (R03HD094560) and the National Institute of Neurological Disorders and Stroke (R01NS100952 and R41NS103698) of the NIH. W.P.M. has received royalties from ABC-Clio Publishing, Springer International, and Wolters Kluwer and research support from the National Hockey League Alumni Association through the Corey C. Griffin Pro-Am Tournament and the National Football League. T.A.B. has received funding related to this research from the National Institute of Neurological Disorders and Stroke of the NIH (R03NS104371) and the CARE Consortium of the National Collegiate Athletic Association and the Department of Defense (W81XWH- 14-2-0151). AOSSM checks author disclosures against the Open Payments Database (OPD). AOSSM has not conducted an independent investigation on the OPD and disclaims any liability or responsibility relating thereto.

before analysis but not during enrollment, as data collection took place before publication of the traumatic encephalopathy syndrome criteria.[26] Additionally, a current concussion at the time of testing, as well as any current or previous lower extremity orthopaedic injury or neurological, visual, or vestibular condition that would interfere with normal gait performance, resulted in exclusion from the study. The institutional review board from each university approved all study procedures before recruitment, and all participants provided consent before their participation.

Career Duration

Career duration at one institution was directly self-reported by the participants. The other institution self-reported eAFE to collision sports instead of overall career duration. Therefore, we determined career duration by subtracting the participant's eAFE from his age at the time of testing, as done in the eAFE literature.[11-14] Despite the difference in career duration metrics between testing sites, the results were not significantly different when each site was analyzed independently. In addition to career duration and/or eAFE, participants had to self-report the number of diagnosed concussions sustained before testing.[18]

Instrumentation

Participants wore 3 wearable inertial measurement units (Opal Sensors; APDM Inc) attached to the dorsal surface of each foot and along the spine at the level of the L5 vertebra. The inertial measurement units are feasible for use in a clinical setting, reliable, and valid for use in neurologically impaired and healthy populations.[5,28-30] Each inertial measurement unit included a triaxial accelerometer, gyroscope, and magnetometer and sampled at 128 Hz.

Procedures

Gait assessments were included as a component of each participant's preseason baseline concussion testing. The gait assessment consisted of participants performing 5 gait trials of a single-task and a dual-task condition. All participants began each gait trial standing with feet together behind a visible starting line and, in response to a verbal cue, walked down a 10-m walkway at a self-selected pace, turned around a specified endpoint, and returned to the starting line.

During dual-task trials, participants performed the same gait protocol while performing a cognitive screening examination of working memory questions: reciting the months of the year in reverse order, spelling 5-letter words backward, and serial subtraction by 6s or 7s from a randomly selected number.[21,23] The instructions were to begin walking when the first question was administered. No instructions were given regarding prioritizing the cognitive or motor task, and the order of the cognitive tasks was randomized.

Data and Statistical Analysis

This cross-sectional study examined the relationship between career duration of collision sports and single- and dual-task gait performance. The independent variable was career duration, which was calculated as the athlete's age minus his eAFE and analyzed as a continuous variable. Concussion history, which was analyzed as a dichotomous variable (yes/no), and height were included as covariates. The dependent gait variables (single- and dual-task gait speed and stride length) were measured using the inertial measurement units, calculated using Mobility Lab software,[29] and then averaged to provide an average value for each trial. Gait speed was reported as the distance divided by time, and the stride length was calculated from the distance between consecutive heel strikes on the same foot and analyzed from the angular velocities recorded using the gyroscopes on the left and right feet.[2,29] We also calculated the motor dual-task cost for gait speed and stride length as the percentage change between the single- and dual-task conditions: (dual – single)/single.[22] Negative dual-task cost variables were indicative of worse motor performance (slower gait speed, shorter stride length) under dual-task versus single-task conditions. The cognitive dependent variable was cognitive accuracy (the percentage of correct answers out of the total number of possible answers given during the dual-task trials). The mean of the 5 trials was used for each variable.

We compared the separate site characteristics using independent-samples t tests. To examine the relationship between career duration and gait performance, combined and at the 2 sites individually, we conducted a series of linear regression models for each outcome measure while adjusting for the potential effect of concussion history and height. Eta-squared effect sizes were used to interpret clinically meaningful results, with 0.02, 0.13, and 0.26 corresponding to small, medium, and large effect sizes.[38] All statistical models were performed using Stata/IC 15.1 (StataCorp). The alpha level was <0.05.

RESULTS

There were 171 male student-athletes eligible for participation in our study. Three participants were excluded for having a career duration <5 years, leaving 168 individuals included in our analyses. Most participating athletes (58%) played football, and approximately a quarter had a history of sport-related concussion. The career duration ranged from 5 to 19 years, and there was no difference in career duration between sites (Table 1). The results of the site-specific analyses can be found in the Appendix (available in the online version of this article).

Gait Speed

There was no significant association between career duration and single-task gait speed ($\beta = -0.004$; $P = .35$; 95% CI = -0.012 to 0.004; $\eta^2 = 0.005$) or dual-task gait speed

TABLE 1
Participant Characteristics for the Combined and Separate Sites[a]

Variable	Combined (N = 168)	Site 1 (n = 138)	Site 2 (n = 30)	*P* Value
Age, y	19.2 ± 1.3	19.4 ± 1.3	18.3 ± 0.7	<.001[b]
Height, cm	184.5 ± 7.2	184.4 ± 7.1	185.3 ± 7.7	.54
Mass, kg	94.3 ± 15.9	93.6 ± 15.5	97.2 ± 17.8	.26
eAFE, y	8.6 ± 3.1 (2-17)	8.7 ± 3.1 (2-17)	8.1 ± 2.9 (3-13)	.33
Career duration, y	10.6 ± 3.0 (5-19)	10.7 ± 3.0 (5-19)	10.3 ± 3.0 (5-15)	.51
Concussion history, No. (%)				
No	124 (74)	101 (73)	23 (77)	
Yes	44 (26)	37 (27)	7 (23)	
Sport breakdown, %				
Football	58	54	77	
Lacrosse	39	46	6.5	
Ice hockey	2		10	
Wrestling	1		6.5	

[a]Values are presented as mean ± SD (range) where appropriate. eAFE, estimated age of first exposure. The blank cells indicate that there were no ice hockey or wrestling athetes from Site 1 or no statistical analyses done.
[b]There was a significant difference between the sites for age.

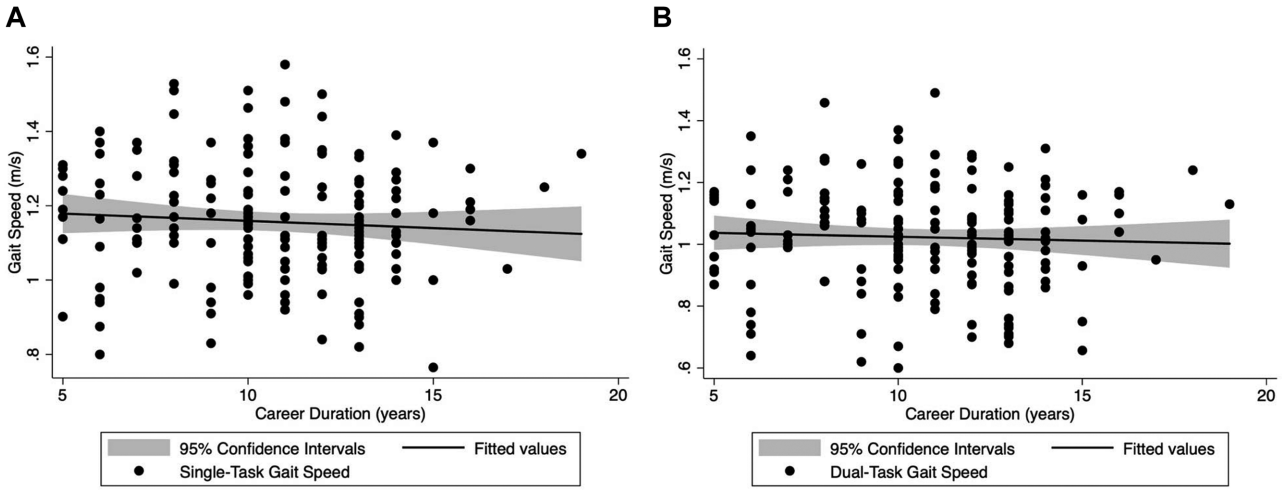

Figure 1. (A) Career duration and single-task gait speed. There was not a significant association between career duration and single-task gait speed (mean ± SD, 1.16 ± 0.16 m/s; $\beta = -0.004$; $P = .35$; $R^2 = 0.005$). (B) Career duration and dual-task gait speed. There was not a significant association between career duration and dual-task gait speed (mean ± SD, 1.02 ± 0.17 m/s; $\beta = -0.003$; $P = .57$; $R^2 = 0.002$).

($\beta = -0.003$; $P = .57$; 95% CI = −0.011 to 0.006; $\eta^2 = 0.002$) (Figure 1, Table 2). Additionally, there was no significant association between career duration and gait speed dual-task cost (mean ± SD; −11.63% ± 7.97%; $\beta = 0.069$; $P = .73$; 95% CI = −0.339 to 0.478; $\eta^2 = 0.0007$).

Stride Length

There was no significant association between career duration and single-task stride length (1.27 ± 0.13 m; $\beta = -0.005$; $P = .09$; 95% CI = −0.012 to 0.001; $\eta^2 = 0.017$) (Figure 2A). Similarly, there was no significant association between career duration and dual-task stride length (1.19 ± 0.14 m; $\beta = -0.006$; $P = .12$; 95% CI = −0.013 to 0.002; $\eta^2 = 0.014$) (Figure 2B, Table 2). There was also no statistically significant association between career duration and stride length dual-task cost (−6.79% ± 4.80%; $\beta = -0.047$; $P = .71$; 95% CI = −0.294 to 0.199; $\eta^2 = 0.0009$).

Cognitive Accuracy

There was no statistically significant association between career duration and cognitive accuracy during dual-task walking ($\beta = -0.004$; $P = .21$; 95% CI = −0.010 to 0.002; $\eta^2 = 0.010$) (Figure 3).

DISCUSSION

We sought to determine if a longer career duration of collision sports and therefore greater potential exposure to RHI

TABLE 2
Gait Speed and Stride Length Data by Career Duration[a]

		Gait Speed, m/s		Stride Length, m	
Duration, y	No.	Single Task	Dual Task	Single Task	Dual Task
5	9	1.18 ± 0.13 (0.91-1.31)	1.03 ± 0.12 (0.87-1.17)	1.31 ± 0.13 (1.12-1.50)	1.21 ± 0.13 (1.05-1.45)
6	12	1.12 ± 0.21 (0.80-1.40)	0.97 ± 0.22 (0.64-1.35)	1.26 ± 0.16 (1.00-1.49)	1.16 ± 0.18 (0.90-1.40)
7	9	1.18 ± 0.12 (1.02-1.37)	1.09 ± 0.11 (0.99-1.24)	1.30 ± 0.10 (1.14-1.42)	1.25 ± 0.09 (1.12-1.38)
8	13	1.26 ± 0.16 (0.99-1.53)	1.12 ± 0.16 (0.88-1.46)	1.34 ± 0.14 (1.13-1.60)	1.27 ± 0.12 (1.06-1.46)
9	10	1.11 ± 0.18 (0.83-1.37)	0.96 ± 0.20 (0.62-1.26)	1.22 ± 0.16 (0.95-1.46)	1.13 ± 0.18 (0.80-1.41)
10	28	1.18 ± 0.15 (0.96-1.51)	1.03 ± 0.17 (0.60-1.37)	1.31 ± 0.11 (1.15-1.54)	1.21 ± 0.12 (0.91-1.43)
11	16	1.17 ± 0.20 (0.92-1.58)	1.05 ± 0.19 (0.79-1.49)	1.28 ± 0.14 (1.10-1.55)	1.22 ± 0.15 (1.02-1.51)
12	19	1.16 ± 0.17 (0.84-1.50)	1.01 ± 0.16 (0.70-1.29)	1.28 ± 0.14 (1.11-1.60)	1.19 ± 0.15 (0.90-1.54)
13	27	1.09 ± 0.14 (0.82-1.34)	0.96 ± 0.16 (0.68-1.25)	1.20 ± 0.10 (1.03-1.39)	1.11 ± 0.12 (0.90-1.31)
14	13	1.16 ± 0.11 (1.00-1.39)	1.05 ± 0.14 (0.86-1.31)	1.30 ± 0.11 (1.13-1.46)	1.22 ± 0.14 (0.96-1.41)
15	5	1.06 ± 0.23 (0.77-1.37)	0.92 ± 0.21 (0.66-1.16)	1.20 ± 0.15 (0.99-1.39)	1.13 ± 0.15 (0.90-1.26)
16	4	1.22 ± 0.06 (1.16-1.30)	1.12 ± 0.06 (1.04-1.17)	1.24 ± 0.08 (1.13-1.30)	1.16 ± 0.10 (1.01-1.23)
17	1	1.03	0.95	1.15	1.12
18	1	1.25	1.24	1.33	1.31
19	1	1.34	1.13	1.20	1.07

[a]Values are presented as mean ± SD (range) where appropriate.

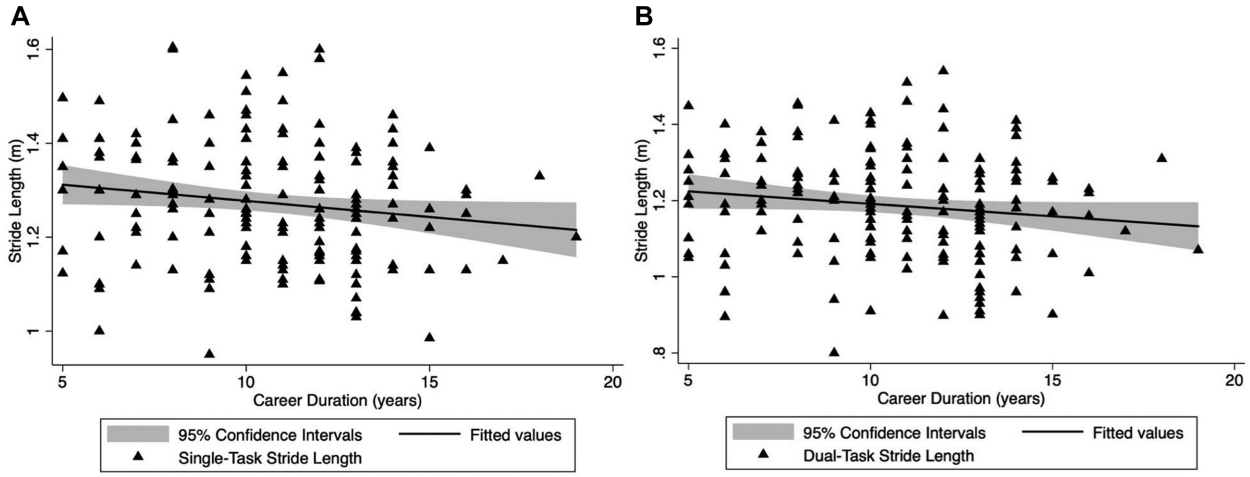

Figure 2. (A) Career duration and single-task stride length. There was no significant association between career duration and single-task stride length (mean ± SD, 1.27 ± 0.13 m; $\beta = -0.005$; $P = .09$; $R^2 = 0.101$). (B) Career duration and dual-task stride length. There was no significant association between career duration and dual-task stride length (mean ± SD, 1.19 ± 0.14 m; $\beta = -0.006$; $P = .12$; $R^2 = 0.052$).

impaired gait performance in collegiate athletes. We observed no statistically significant associations between career duration and single/dual-task gait performance. Additionally, there were no associations between career duration and cognitive accuracy or motor dual-task costs. Thus, the overall results of this study suggest no detrimental effects of career duration on single- or dual-task gait performance or dual-task cognitive performance in collegiate student-athletes; however, the later life effects remain to be elucidated.

Career duration has largely been examined in the context of later life outcomes (traumatic encephalopathy syndrome, chronic traumatic encephalopathy) in retired or deceased former collision sport athletes.[15,36,37,45] Longer career duration has been associated with an increased number of reported concussions,[15] increased odds of developing chronic traumatic encephalopathy,[37] and a diagnosis of chronic traumatic encephalopathy at death.[36] Conversely, when using age of retirement as a marker of career duration, authors have found no association with

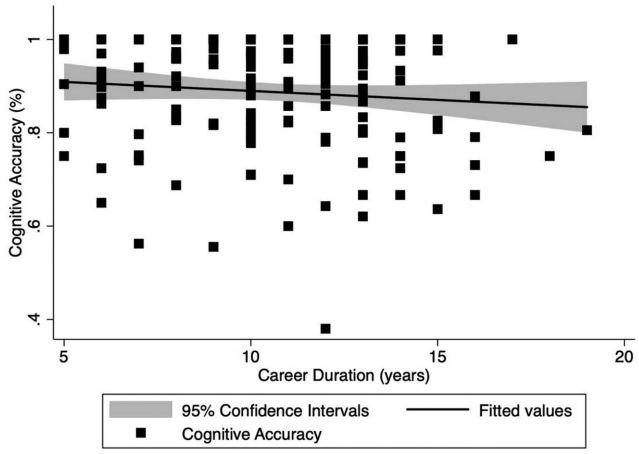

Figure 3. Career duration and cognitive accuracy. There was no significant association between career duration and cognitive accuracy (mean ± SD, 89% ± 12%; β = −0.004; P = .21; R^2 = 0.015).

the pathologic presence of chronic traumatic encephalopathy upon autopsy.[45] We observed no significant associations between single- or dual-task gait speed and collision sport career duration, and gait speeds fell within the normal range for healthy collegiate student-athletes (single task, 1.16 m/s; dual task, 1.02 m/s).[21] Likewise, the stride lengths in our population (single task, 1.27 m; dual task, 1.19 m) were nearly identical to the mean stride lengths previously observed for healthy collegiate student-athletes participating in collision sports.[21] Therefore, a longer career duration (greater exposure to RHI) was not associated with gait performance among current collegiate athletes playing collision sports.

While career duration and eAFE are different markers, both are indicators for the level of exposure to RHI over time. Our findings support recent investigations from Caccese et al,[11-14] who found no association between earlier exposure to RHIs and worse neurocognitive function, psychological distress, symptoms, symptom recovery, or postural stability in a similar population of healthy NCAA athletes. Although these previous studies were able to examine concussion outcomes from thousands of NCAA athletes, the outcomes were measured using clinical assessments with known limitations. They evaluated postural stability using the Balance Error Scoring System, which has lower sensitivity than other dynamic tasks, such as instrumented and tandem gait.[7,41] However, the similarity of our findings using measures of gait, which is one marker of central nervous system health,[17,43] provides additional evidence that neither quiet stance nor dynamic postural control tasks are negatively affected by longer exposure to RHI in college-aged individuals. As a marker of neurological health, gait speed is considered an indicator of functional and physiological disturbances,[17] and dual-task gait speed is an important element of neurological functioning given its ability to be used to evaluate motor and cognitive function simultaneously. Recent evidence has also indicated that at 45 years old, gait speed is associated with physiological indicators of accelerated aging, lifelong compromised brain health, and neurocognitive function.[43] The absence of an association between gait speed and career duration in this investigation demonstrates that longer exposure to RHI does not appear to manifest in gait-based neurological impairment significant enough to cause diminished postural control performance this early in life but warrants further investigation across the life span.

In addition to our limited findings on gait outcomes, we did not observe any significant associations between career duration and either cognitive accuracy during dual-task gait or motor dual-task costs (gait speed and stride length). Greater dual-task costs have been associated with poorer outcomes (ie, prolonged concussion symptoms,[19] time-loss injuries after a concussion[20]), but the absence of significant and clinically relevant findings with dual-task motor and cognitive variables within this investigation suggest that longer exposure to RHI does not result in gait-based neurological deficits in college-aged collision sport athletes. Greater dual-task costs for stride time variability have been observed in retired professional football players, although these findings were independent of career duration.[31] The literature surrounding the association of eAFE and career duration with cognitive function remains mixed. A number of studies specific to football have suggested that earlier exposure to tackling (ie, before age 12 years) may result in greater cognitive impairments later in life.[1,44,47,48] Other studies that have examined athletic populations in addition to football found no such relationship at the collegiate level and beyond.[11-14,46] A series of recent studies in a similar population of NCAA athletes revealed no evidence of an association between eAFE and neurocognitive function, but it remains to be seen how the consequences of RHIs play out across the life span.[11-14] Furthermore, only 28% of the nearly 50 studies included in a recent systematic review of long-term clinical outcomes in retired athletes indicated greater cognitive impairments in athletes with more exposure to head impacts,[16] suggesting that there are more risk factors at play than just RHIs when it comes to potential long-term damage from collision sports.

One limitation of our study was that career duration was based on participant's self-reported eAFE to collision sports. This approach, though, is consistent with that of previous eAFE studies investigating similar populations,[11-14,25] as well as studies of older/retired athletes.[25,44,47,48] We were also unable to verify that individuals participated in collision sports continuously from the age that they self-reported, but as both institutions were sponsored NCAA Division I intercollegiate athletic programs, it is highly likely that this approach is accurate, as lengthy breaks in participation would be quite unusual. Furthermore, we were unable to account for rule differences across our participants' collision sport history (eg, a participant may have played hockey for 10 years but was allowed full contact/checking for just 8 of those years), so frequent collisions may not have occurred for the entire career duration. Our study examined male collision sport athletes at the collegiate level, as there were no female participants in football or wrestling and women's

lacrosse and ice hockey are not collision sports. Therefore, we are unable to extrapolate these findings to female athletes in sports that involve RHIs (eg, soccer, rugby). This study used a cross-sectional design, so we cannot draw any conclusions as to how earlier exposure to collision sports may affect an individual across the life span. It is possible that postural control impairments from RHI are not significant enough to manifest at the collegiate age, so future longitudinal studies should be conducted to track athletes across the life span.

In summary, current collegiate collision sport athletes demonstrated no significant differences in single- or dual-task gait based on their career duration participation in collision sports. Since gait is one marker of neurological health, our lack of significant findings leads us to believe that a longer career duration of collision sports is not detrimental to gait-based central nervous system health, at least by college age. However, the neurological cost of diminished gait performance later in life remains to be seen and should be a topic of future studies.

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